

Reporting checklist for deployment-matched structural ML validation

Each item forces the validation score to identify what kind of generalization it supports.

01	Target	State the intended deployment target within-source, unseen source, unseen mix, unseen section family, time block, spatial block
02	Grouping	Define the grouping variable before modelling source reference, experimental program, mixture family, author/source family, section family
03	Random baseline	Report the conventional random split score so readers can compare with prior literature
04	Grouped validation	Report a target-matched grouped split GroupKFold, leave-one-source-out, leave-one-family-out
05	Topology	Report group topology diagnostics group count, median size, singleton rate, imbalance
06	Classification balance	Report class balance within groups required for failure-mode classification
07	Regression metrics	Report pooled and fold-level regression metrics pooled R2 plus fold-level diagnostics; MAE/RMSE when available
08	Uncertainty	Attach uncertainty or sensitivity diagnostics bootstrap, fold-summary interval, leave-one-group sensitivity
09	Data provenance	List original data sources and licenses do not treat a PDF repository record as raw data
10	Code release	Release scripts and environment files download, preprocessing, validation, figures
11	Failure log	Report failed or partial reproductions embargo, metadata-only, no source labels, blocked access
12	Claim discipline	Match claims to the validation target do not call within-source interpolation out-of-source generalization

Use this checklist with random and grouped validation scores; a high random score alone is not evidence of out-of-source generalization.